Learning how to Prove Workshop Week 6

Groups

Groups and subgroups

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1. **Show that there are only two groups of order 4, the cyclic group C_4 and a noncyclic group K_4,** whose Cayley table is as follows:

K_4	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

2. Let H and K be subgroups of a group G.
   (a) Prove that $H \cap K$ is the largest subgroup of G contained in both H and K. In other words, if L is a subgroup of G such that $L \subseteq H$ and $L \subseteq K$ then $L \subseteq H \cap K$.
   (b) Show that $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$.